

COM3105 Advanced Algorithms

Homework 3

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Question 1

(4 points)

Consider a stream $a_1, \dots, a_m$ where each $a_i \in \{1, \dots, n\}$.

Give a deterministic streaming algorithm that maintains the sum $\sum a_1 + \dots + a_m$ using $\mathcal{O}(\log m + \log n)$ space.

Answer

answer here

Question 2

(6 points)

Consider a stream $a_1, \dots, a_m$ where each $a_i \in \{1, \dots, n\}$.

Give a randomised streaming algorithm which approximates the sum $\sum a_1 + \dots + a_m$ using $\mathcal{O}(\log \log m + \log n)$ space.

Give a rigorous analysis of the algorithm in terms of its quality and space complexity.

Answer

answer here
